DD Robocon India 2026: MathWorks Modeling Award

Event Name	DD Robocon India 2026: MathWorks Modeling Award
MathWorks Point-of-Contact	Manavendra Desai-Deodhar (mdesaide@mathworks.com)
Date of Announcement	April 2026

A. Award overview

The MathWorks Modeling Award is designed to recognize the teams that demonstrate the best use of MathWorks tools in DD Robocon India 2026. Winners will be selected through a two-phase evaluation process, details of which can be found in Section B.

The teams securing **first**, **second**, and **third** place will receive cash prizes of INR 35000, INR 20000, and INR 15000, respectively. These prizes will be conferred during the DD Robocon India 2026 award ceremony.

Important Note: As per the Income Tax Act 1961, MathWorks India is required to deduct Tax at source from this token of our appreciation. This Tax deduction will be at the rate of 30%. The amount of Tax deducted at Source will be available in your PAN number and can be utilized against your Tax liability for the year or otherwise can be claimed as a refund from the Government of India by timely filling of the Income Tax return for the year.

B. Procedure for participation

- Participation in this competition is free.

- **Phase 1:**
  - The teams must submit their entries by **11:30 p.m. IST on June 14, 2026.** Only one entry can be submitted by each team. The required contents and format of this submission are provided in Sections C and D. The teams must use the form provided in Section E to submit their entry.
  - All entries will be judged by MathWorks engineers on a 100-point scale. The distribution of points is provided in Section C.
  - The result for Phase 1 will be announced on **June 18, 2026**.
  - No more than ten teams will qualify for Phase 2 of the MathWorks Modeling Award.
  - ***A team must be shortlisted for Stage-2 of DD Robocon India 2026 to be eligible for Phase 2 of the MathWorks Modeling Award.***

- **Phase 2:**
  - The teams will be invited to deliver a formal presentation of their work to the MathWorks judges**.** The dates and a WebEx link for this *online* presentation will be sent to the qualified teams on **June 18, 2026**.
  - During the online presentation, the teams will use the report they submitted in Phase 1, and are expected to keep relevant drawings, simulations and/or demonstrations prepared, to facilitate the Q&As at the end of their presentation.

- The teams securing **first**, **second**, and **third** place, will be announced during the award ceremony of DD Robocon India 2026.

- **The judges' decisions are final.**

C. Content for Phase 1 submission

Points **(100 available)** will be awarded to submissions in Phase 1 based on the criteria given below.

(I) Application of Model-Based Design (15 pts):

a) Please describe how the team has implemented core principles of Model-Based Design using MathWorks tools to address the problem statement of the DD Robocon India 2026 competition.
b) Summarize the role that MathWorks tools played in achieving the objectives of the DD Robocon India 2026 competition.
c) Define and explain how MathWorks tools were integrated into each stage of the development process, including
    1. Requirements analysis
    2. Designing through modelling and simulation
    3. Verification and Validation.

Learn about Model-Based Design here: https://www.mathworks.com/solutions/model-based-design .

(II) **Utilization of MathWorks tools for robot development (75 pts):** In relation to the problem statement of DD Robocon India 2026, please describe how the team has utilized MathWorks tools for

a) **Modeling the robot (15 pts):**
    1. Development of models for different components of the robot
    2. Validation of models developed, and their degree of correlation with real-world behavior
    3. Steps taken in designing the models to ensure usability, clarity, extensibility, and efficient execution

Example: Modeling mechanisms and actuators in the robot, with appropriate kinematics and dynamics included, using Simscape Multibody.

b) **Image processing and computer vision (10 pts):** Describe the image processing and computer vision methodologies developed to facilitate decision-making and navigation of the robot

Example: Object-detection using convolution neural networks in the Computer Vision Toolbox.

c) **Path-planning and navigation (15 pts):**
    1. Describe the decision-making logic developed for different actions performed by the robot
    2. Describe the algorithms implemented and/or developed for robot navigation and how data from the image-processing and computer-vision methodologies was used.

Example: Programming the decision-making logic using Stateflow and using the RRT algorithm in the Robotics Systems Toolbox for collision-free path-planning and navigation.

d) **Control Design and Tuning (10 pts):**
    1. Describe the control algorithms utilized and/or developed to control the robot and its components
    2. Describe the plant models used in developing these controllers
    3. Describe the method used for tuning these controllers

4. Describe how the plant models and control algorithms were validated, and the level of correlation between simulated and real-world performance of the robot
5. Describe the steps taken in designing the plant models to ensure usability, clarity, extensibility, and efficient execution of the control algorithm.

Example: Creating a linear model for your control algorithm using the Control System Toolbox, and Tuning a PID controller using the PID Tuner app.

e) **Data Analysis (10 pts):** Describe how data on the deployed robot is acquired, analyzed, and visualized.

Example: Acquiring and visualizing sensor data from the robot on a dashboard created using the MATLAB App Designer.

f) **Embedded systems and deployment on hardware (15 pts):**
1. Describe how the models and algorithms, developed and tested in simulation, were deployed on the physical robot
2. Describe how the automatic code-generation capability in MATLAB and Simulink was used in deploying models and algorithms on the physical robot and its embedded hardware
3. Explain how the physical robot is begin tested and tuned for the DD Robocon India 2026 competition.

Example: Using MATLAB Coder to automatically convert functions of the Robotics System Toolbox into C/C++ code for deployment on hardware.

(III) **Quality, presentation, and readability, of information, schematics, and code snippets, in your submission (5 pts)**.

(IV) **A brief discussion on your plans for next year based on the lessons already learnt this year (5 pts).**

Example: What would you do differently next time and what more would you like in the current MathWorks tools you have access to.

D. Format for Phase 1 submission

Your entry must be in the form of a report. The report **should not be more than five pages long** and the text must follow a **two-column layout**. The report must be **submitted as a PDF** with a '.pdf' file extension.

E. Phase 1 submission

Submit your entry for Phase 1 of the MathWorks Modeling Award using this Microsoft Form: Report Submission for DD Robocon India 2026 - MathWorks Modeling Award

Please name your entry using the following format: *TeamName_MathWorks*_2026_*Version_#*. For example: *TeamRobotSquad_MathWorks_2026_Version_1*.

To avoid download errors, please submit the file as a PDF with a '.pdf' file extension.

The deadline for submitting your entry is **11:30 p.m. IST on June 14, 2026.**

F. Resources to get started

Recommended self-paced tutorials

- MATLAB Onramp
- Simulink Onramp
- Stateflow Onramp

Modeling physical systems

- Simulating Pneumatic Robot Actuators
- Simulating Robot Throwing Mechanisms
- Control Design for Robot Throwing Systems
- Model and Control a Manipulator Arm with Robotics and Simscape - MATLAB & Simulink
- Design and Simulate Warehouse Pick-and-Place Application Using Mobile Manipulator in Simulink and Gazebo - MATLAB & Simulink
- GitHub - mathworks/Industrial-Robots-Simscape: Manipulator and mobile robot models for trajectory planning and actuator analysis. · GitHub

Practical approaches to robot perception

- Basic Operations on Images
- Image Segmentation and Analysis
- Feature Matching and Tracking
- MATLAB and Simulink ROS Tutorials
- Mapping for Mobile Robots and UGV
- 'How To' – Videos using MATLAB and Simulink

Robot manipulation and locomotion

- Robot Manipulators - MATLAB & Simulink
- Perform Pick and Place with Collision-Object-Based Obstacle Avoidance in Robot Scenario - MATLAB & Simulink
- MATLAB and Simulink Robotics Arena: Walking Robot - File Exchange - MATLAB Central
- Running Robot Model in Simscape - File Exchange - MATLAB Central

Navigation

- Mobile Robots - MATLAB & Simulink
- Obstacle Avoidance with TurtleBot and VFH - MATLAB & Simulink
- A* Path Planning and Obstacle Avoidance in a Warehouse - MATLAB & Simulink
- Perform Obstacle Avoidance in Warehouse Scenario with Mobile Robots - MATLAB & Simulink

Code generation

- Tutorials and Videos - MATLAB & Simulink

Published Videos/Blogs by Teams

- DD ROBOCON India 2025 MathWorks Modeling Award: From Simulation to the Real-World with BRACT's VIT Pune » Student Lounge - MATLAB & Simulink
- Rescue on Mars: RTC Gransee team's journey in the RoboCup Junior Onstage
- Climb stairs and shoot the target: A Student Robotics Project!
- Designing Arrow Throwing Robots by Team DJS Robocon
- Designing Robots to Play Rugby by Team KJSCE Robocon
- Team Automatons Builds a Quadruped Robot for the ABU Robocon 2019 Task

General resources

- Robotics and Autonomous Systems - MATLAB & Simulink
- GitHub - mathworks-robotics/awesome-matlab-robotics: This is a list of awesome demos, tutorials, utilities and overall resources for the robotics community that use MATLAB and Simulink. · GitHub